

The Space Cost of Shor’s Algorithm: How Large Must the Quantum Phase Estimation Register Be?

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Abstract

Shor’s algorithm reduces integer factoring to order-finding via Quantum Phase Estimation (QPE). While the target register that holds the number N needs only $n = \lceil \log_2 N \rceil$ qubits, the counting register that reads the phase must be substantially larger to resolve it with enough precision. This work quantifies that overhead empirically: by sweeping the counting register size t for $N \in \{15, 21, 35, 105\}$ and measuring the probability of resolving the phase and of successfully factoring N , we find that the smallest reliable counting register is $t^* \approx 1.5\text{--}1.6n$, comfortably inside the worst-case bound $t \leq 2n + 1$, and that this phase-precision cost dominates the total circuit width.

1 Shor’s Algorithm: the Full Pipeline

Factoring N reduces to finding the order r of a random a coprime to N : $\gcd(a^{r/2} \pm 1, N)$ yields a non-trivial factor with probability $\geq 1 - 1/2^{k-1}$, where k is the number of distinct prime factors of N . The pipeline has four classical/quantum stages:

1. **Order-finding reduction** of factoring to finding r .
2. **Quantum Phase Estimation**, which estimates the eigenphase $\theta = k/r$ of the modular multiplication operator U_a on a counting register of t qubits.
3. **Continued fractions**: the measured phase $y/2^t$ is expanded to recover the best rational approximation k/r with denominator $< N$.
4. **Euclidean algorithm**: once r is known, $\gcd(a^{r/2} - 1, N)$ is computed classically in polynomial time.

2 Inside the QPE Circuit

The operator $U_a |x\rangle = |ax \bmod N\rangle$ is a permutation, hence unitary, with eigenvectors $|u_j\rangle = \frac{1}{\sqrt{r}} \sum_k \omega_r^{-jk} |a^k \bmod N\rangle$, $j = 0, \dots, r-1$, and eigenvalues $\omega_r^j = e^{2\pi i j/r}$. The starting state $|1\rangle = \frac{1}{\sqrt{r}} \sum_j |u_j\rangle$ is an equal superposition of *all* eigenvectors of U_a : it requires no prior knowledge of r , unlike preparing a single $|u_j\rangle$ directly.

Order finding therefore uses two registers (Fig. 1): a **target register** T of $n = \lceil \log_2 N \rceil$ qubits holding $|a^k \bmod N\rangle$, and a **counting register** C of t qubits that reads the phase $\theta = k/r$ via controlled modular multiplications followed by an inverse QFT.

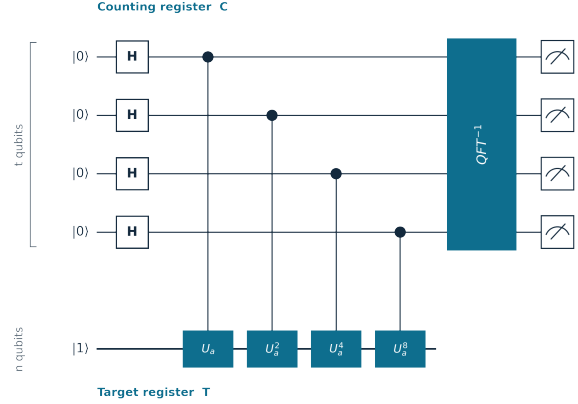


Figure 1: QPE circuit for order finding: the counting register C (t qubits) controls modular multiplications M_b on the target register T (n qubits), followed by an inverse QFT and measurement.

3 The Space Cost: $t \approx 2n$

Measuring C returns $y/2^t \approx k/r$ with k uniform in $\{0, \dots, r-1\}$. Continued fractions reconstruct k/r exactly only if the phase is resolved with error $< 1/(2r^2)$. Since $r < N$, it suffices to require

$$\left| \frac{y}{2^t} - \frac{k}{r} \right| < \frac{1}{2N^2} \implies t \gtrsim 2 \log_2 N + 1 \approx 2n,$$

i.e. the counting register that only reads the phase must be roughly *twice as large* as the register holding the number itself. The total circuit width is therefore $\approx 3n$ qubits, against $\log_2 N$ bits of useful information extracted.

4 Experimental Setup

We fix the target $N \in \{15, 21, 35, 105\}$ with base $a = 2$ and sweep only t , simulating the exact distribution of register C (PennyLane default.qubit, analytic mode, no sampling shots). Two probabilities are measured for each (N, t) :

- $P_{\text{resolve}}(t)$ — the phase is measured within $1/(2r^2)$ of a true k/r : a purely spatial metric (“did the register have enough bits?”).
- $P_{\text{factor}}(t)$ — a single shot yields a non-trivial factor of N : it additionally accounts for the numerical overhead from $k = 0$, odd r , or $\gcd(k, r) > 1$.

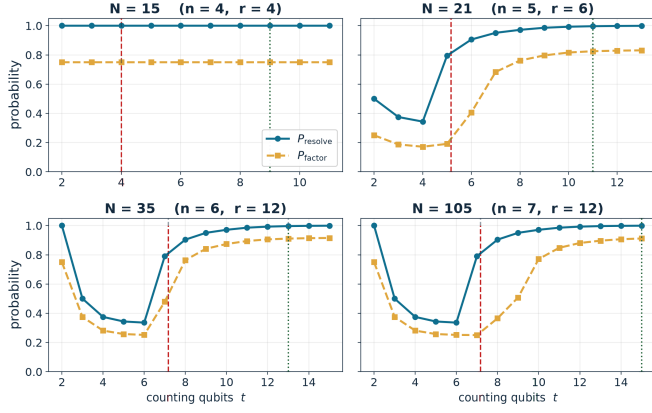


Figure 2: P_{resolve} (circles) and P_{factor} (squares) vs. counting-register size t , for $N = 15, 21, 35, 105$. Dashed line: $t = 2 \log_2 r$; dotted line: worst-case bound $t = 2n + 1$.

The gap between the two is fixed by Shor’s theorem ($P_{\text{factor}} \geq 1 - 1/2^{k-1}$), while the point where P_{resolve} rises marks the true space requirement.

5 Results

5.1 Success vs. register size

Fig. 2 shows P_{resolve} and P_{factor} against t for all four values of N . The “knee” where P_{resolve} rises occurs around $t \approx 2 \log_2 r$, well inside the worst-case bound $2n + 1$; P_{factor} then plateaus below 1, reflecting the constant numerical overhead. $N = 15$ is a degenerate case: $r = 4 = 2^2$, so $k/4$ is exactly representable in 2 bits and the curves are flat from the start.

5.2 The non-monotonic dip

For $N = 15$, tiny t can look deceptively successful: with very few bits, the only representable phase values $y/2^t$ coincide by aliasing with true k/r values. As t grows just enough to sample phases that are *not* k/r , but not yet enough to resolve them, success temporarily collapses before recovering once $t \gtrsim 2 \log_2 r$ (Fig. 3).

5.3 Space requirement vs. numerical overhead

Fig. 3 isolates the two contributions: the *rise* of P_{resolve} marks the point where the register reaches $\approx 2 \log_2 r$ bits (the true space cost), while the residual gap to P_{factor} — constant in t — is the numerical overhead from $k = 0$, odd r , or $\gcd(k, r) > 1$, reducible only by repeating the circuit or changing a . We define t^* as the smallest counting-register size for which $P_{\text{factor}}(t)$ reaches 90% of its plateau value.

5.4 Width and circuit cost

Circuit width $t + n$ grows linearly in t , while the inverse QFT’s controlled-phase gate count $t(t - 1)/2$ grows quadratically (Fig. 4). $t^* \approx 1.5\text{--}1.6n$ across the non-degenerate instances ($N = 15$ is the lucky case at $t^*/n \approx 0.5$).

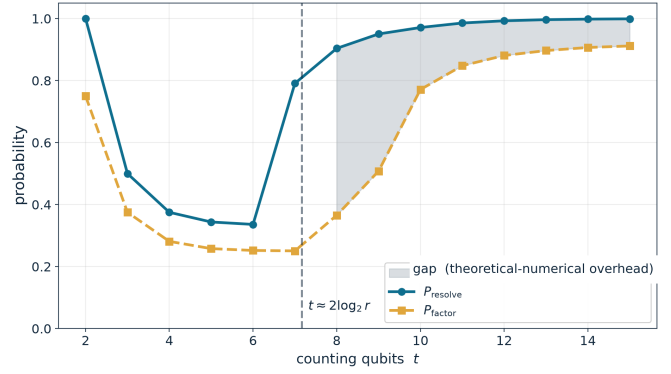


Figure 3: Space requirement (rise of P_{resolve}) vs. numerical overhead (residual gap to P_{factor}), shaded in grey.

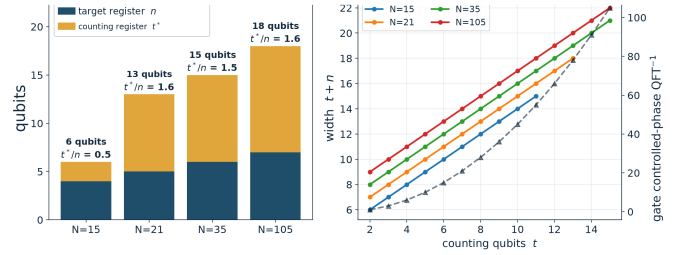


Figure 4: Target vs. counting register width at t^* (left), and circuit width / inverse-QFT gate count vs. counting qubits (right).

N	n	t^*	width	t^*/n
21	5	8	13	1.60
35	6	9	15	1.50
105	7	11	18	1.57

Table 1: Measured space requirement t^* and total circuit width ($t^* + n$) for the non-degenerate instances.

6 Conclusions

The phase-only counting register is empirically $\sim 1.5\times$ the size of the register holding the number itself, consistent with Shor’s asymptotic result $t^* \sim n + m$ (with $P_{\text{factor}} \geq 1 - 1/2^m$), and well within the worst-case bound $t \leq 2n + 1$. Within the same circuit, width grows linearly while the inverse QFT’s gate count grows quadratically, and only a handful of distinct modular-multiplication operators M_b are actually needed. Two-thirds of the total circuit width is spent purely on phase precision rather than on representing N itself — the dominant space cost of Shor’s algorithm on near-term hardware.

References

- [1] P. W. Shor, “Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer,” *SIAM J. Comput.*, vol. 26, no. 5, 1997. arXiv:quant-ph/9508027.
- [2] IBM Quantum, “Shor’s algorithm — tutorial,” <https://quantum.cloud.ibm.com/docs/>

[en/tutorials/shors-algorithm.](#)